



Lab Practical #02:

Study of different network devices in detail.

Practical Assignment #02:

1. Give difference between below network devices.

- Hub and Switch
- Switch and Router
- Router and Gateway

2. Working of below network devices:

- Switch
- Router
- Gateway

Hub and Switch

No.	Hub	Switch
1	Operates at Physical Layer (Layer 1) of OSI model.	Operates at Data Link Layer (Layer 2) of OSI model.
2	Broadcasts data to all ports regardless of destination (flooding).	Forwards data only to the specific destination port using MAC address table.
3	Half-duplex: devices cannot send and receive simultaneously.	Full-duplex: devices can send and receive data at the same time.
4	Creates a single collision domain for all connected devices.	Each port is a separate collision domain, reducing network collisions.
5	No intelligence; does not learn or store MAC addresses.	Maintains a MAC address table (CAM table) to intelligently forward frames.

Switch and Router

No.	Switch	Router
1	Operates at Data Link Layer (Layer 2) of OSI model.	Operates at Network Layer (Layer 3) of OSI model.
2	Uses MAC addresses to forward data frames within a LAN.	Uses IP addresses to route packets between different networks.
3	Connects devices within the same Local Area Network (LAN).	Connects multiple networks including LANs and WANs.
4	Does not maintain a routing table; uses MAC address table only.	Maintains a routing table and uses routing protocols (RIP, OSPF, BGP).
5	All ports share the same broadcast domain.	Each interface is a separate broadcast domain, reducing broadcast traffic.



Router and Gateway

No.	Router	Gateway
1	Routes packets between networks using IP addresses.	Acts as an entry/exit point between networks, often translating protocols.
2	Operates at Network Layer (Layer 3) of the OSI model.	Can operate at multiple OSI layers depending on the gateway type.
3	Works with networks using the same protocol (e.g., IP to IP).	Can interconnect networks using entirely different protocols.
4	Forwards packets based on IP routing table entries.	May perform protocol conversion, data format translation, and encapsulation
5	Typically does not perform Network Address Translation (NAT) by default.	Often performs NAT to map private IP addresses to a public IP address.

Working of below network devices:

1. Switch

A switch is a networking device used to connect multiple devices such as computers, printers, servers, and access points within the same Local Area Network (LAN). It operates mainly at the Data Link Layer (Layer 2) of the OSI model. When a device sends data, the switch receives the frame and examines the destination MAC address. It maintains a MAC address table that stores the addresses of all connected devices along with their corresponding ports. Using this table, the switch determines the correct destination port and forwards the frame only to the intended device. This selective forwarding reduces unnecessary network traffic and minimizes collisions. Unlike a hub, which broadcasts data to all devices, a switch sends data only where it is needed. Modern switches can support high-speed communication, VLANs, and network security features. Switches improve network performance by providing dedicated bandwidth to each connected device. They are commonly used in homes, offices, schools, and data centers. By efficiently managing data traffic, switches help create reliable and fast local networks

2. Router

A router is a network device that connects different networks and enables communication between them. It operates at the Network Layer (Layer 3) of the OSI model and uses IP addresses to route data packets. When a packet arrives at the router, it reads the destination IP address contained in the packet header. The router then consults its routing table to determine the best path for forwarding the packet. If the destination network is directly connected, the router sends the packet there; otherwise, it forwards the packet to another router. Routers continuously exchange routing information to maintain efficient communication paths. In homes and offices, routers connect local networks to the Internet. They also perform functions such as Network Address Translation (NAT), packet filtering, traffic management, and firewall protection. Routers help prevent unnecessary traffic from entering a network and improve overall security. Wireless routers additionally provide Wi-Fi connectivity for mobile devices. Without routers, communication between different networks and Internet access would not be possible.



3. Gateway

A gateway is a networking device or software application that serves as an entry and exit point between different networks. It allows devices on one network to communicate with devices on another network, especially when the networks use different communication protocols. When data needs to leave a local network, it is first sent to the gateway. The gateway receives the data, processes it, and determines where it should be forwarded. If the source and destination networks use different protocols, the gateway translates the data into a compatible format before forwarding it. This protocol conversion capability makes gateways essential in heterogeneous network environments. In most home and office networks, the router acts as the default gateway. Every device on the network is configured with the gateway's IP address so that it knows where to send data destined for external networks. Gateways play a critical role in Internet communication, cloud services, and enterprise networking. They provide connectivity, protocol translation, security, and traffic control. Without gateways, communication between different types of networks would be extremely difficult.